

Incident Report #001 — Brute Force Investigation

Case ID: INC-001

Investigator: SOC Analyst — Lab Investigation

Date: 24 September 2026

Environment: Windows Active Directory Lab

Classification: Controlled Security Lab Simulation

Status: Closed — No Successful Authentication Observed

Severity: Medium (Lab Classification)

1. Executive Summary

On 24 September 2026, a series of failed network authentication attempts was detected against the LAB\soclab domain account on the Windows Server/domain controller WIN-PIVUFUGAE9N.lab.local.

The activity originated from the Kali Linux host at 192.168.100.137 and generated six Windows Security Event ID 4625 records between 19:39:24 and 19:40:54. All six events targeted the same account and originated from the same source address.

The events reported an unsuccessful authentication with status 0xC000006D and sub-status 0xC000006A, consistent with an invalid authentication attempt. The events also recorded Logon Type 3, indicating a network logon, with NTLM listed as the authentication package.

The investigation correlated the failed authentication events with the source IP, workstation name, target account and event timestamps. A separate search for Event ID 4624 using the same target account and source IP returned no matching successful authentication during the investigation window.

Based on the available evidence, the activity is consistent with repeated password-guessing behaviour. However, there is no evidence in this investigation of a successful authentication or subsequent compromise.

The activity was intentionally generated as part of a controlled SOC detection lab. No production systems or real user accounts were involved.

2. Incident Overview

Field	Value
Case ID	INC-001
Detection	Multiple failed authentication attempts
Target System	WIN-PIVUFUGAE9N.lab.local
Target Account	LAB\soclab
Source Host	KALI
Source IP	192.168.100.137
Event ID	4625
Number of Events	6
First Observed	24/09/2026 19:39:24
Last Observed	24/09/2026 19:40:54
Activity Duration	90 seconds
Logon Type	3 — Network
Authentication Package	NTLM
Status	0xC000006D
Sub-status	0xC000006A
Successful 4624 Found	No matching event observed
Environment	Controlled lab
Classification	Suspected password-guessing activity
Final Status	Closed

3. Detection

The detection logic was designed to identify repeated authentication failures against a single account from the same source.

The rule used during the investigation was:

Event ID = 4625

AND

Target Account = LAB\soclab

AND

Source IP = 192.168.100.137

AND

Failed Attempts >= 5

WITHIN 10 minutes

The query returned six matching events.

The threshold was therefore exceeded:

Observed failures: 6

Detection threshold: 5

Detection result: ALERT-001

The alert was treated as an investigation trigger rather than evidence of confirmed compromise.

This distinction is important in a SOC environment. An authentication failure can represent anything from a user entering an incorrect password to automated password guessing. The surrounding context is required before deciding whether the event represents malicious activity.

4. Initial Triage

The initial triage focused on four questions:

1. Which account was being targeted?
2. Where were the authentication attempts coming from?
3. How frequently were they occurring?
4. Was there evidence that any attempt succeeded?

The investigation identified a consistent pattern across all six events.

The target was always:

LAB\soclab

The source was consistently:

192.168.100.137

The workstation recorded in the Windows events was:

KALI

All six events were Event ID 4625 and used Logon Type 3.

The timestamps also showed that the attempts were clustered closely together rather than occurring as isolated user mistakes.

This combination made the activity more suspicious than a single failed login and justified further investigation.

5. Evidence Collected

5.1 Windows Security Events

Six Event ID 4625 records were identified.

Time	Event ID	Record ID	Source IP	Target
19:39:24	4625	125277	192.168.100.137	LAB\soclab
19:40:45	4625	125278	192.168.100.137	LAB\soclab
19:40:47	4625	125279	192.168.100.137	LAB\soclab
19:40:49	4625	125280	192.168.100.137	LAB\soclab
19:40:52	4625	125281	192.168.100.137	LAB\soclab
19:40:54	4625	125282	192.168.100.137	LAB\soclab

The six events span approximately 90 seconds.

5.2 Authentication Details

The detailed event extraction identified the following consistent values:

Event ID: 4625
TargetUserName: soclab
TargetDomainName: LAB
LogonType: 3
WorkstationName: KALI
IpAddress: 192.168.100.137
AuthenticationPackage: NTLM
LogonProcessName: NtLmSsp
Status: 0xC000006D
SubStatus: 0xC000006A

The source ports were different for each event:

57854
48202
48208
48218
48228
48230

This is consistent with separate network connection attempts from the same source rather than one repeated record.

5.3 Authentication Outcome

A separate search was performed for Event ID 4624 using:

Target account = soclab

Source IP = 192.168.100.137

No matching successful authentication event was returned.

Therefore, based on the investigated event data, there is no evidence that the observed authentication attempts resulted in a successful login.

6. Timeline

Time	Event	Investigation Significance
19:39:24	4625 / Record 125277	First observed failed authentication
19:40:45	4625 / Record 125278	Second failed authentication
19:40:47	4625 / Record 125279	Repeated failure
19:40:49	4625 / Record 125280	Repeated failure
19:40:52	4625 / Record 125281	Detection threshold exceeded
19:40:54	4625 / Record 125282	Sixth failure
Investigation 4624 correlation		No matching successful authentication observed

The concentration of six failures within approximately 90 seconds was the main reason the activity was escalated from a simple authentication failure to a security investigation.

7. Indicator of Compromise / Investigation Indicators

The following indicators were relevant to the investigation:

Indicator	Value	Type
192.168.100.137	Kali source address	Source IP
KALI	Source workstation	Hostname
LAB\soclab	Target account	Account
WIN-PIVUFUGAE9N.lab.local	Target system	Host
4625	Failed logon	Windows Event ID
0xC000006D	Authentication failure status	Status
0xC000006A	Invalid credentials sub-status	Sub-status

These indicators should be treated as **lab-specific investigation indicators**, not as evidence that the IP address or account is malicious in a real environment.

8. MITRE ATT&CK Mapping

Tactic

Credential Access — TA0006

Technique

T1110 — Brute Force

Sub-technique

T1110.001 — Password Guessing

The observed behaviour is most closely aligned with password guessing because repeated authentication attempts were made against the same account and the events indicate unsuccessful credential validation.

MITRE ATT&CK describes T1110.001 as the repeated or systematic guessing of passwords to attempt access to an account. MITRE's current detection strategy for password guessing also includes correlation of repeated Windows Event ID 4625 authentication failures over a defined time window.

The evidence does **not** support classifying this event as password spraying because the observed activity was directed at one account rather than multiple accounts. Password spraying is specifically characterised by attempting a password or small set of passwords against many different accounts.

9. Investigation Findings

The investigation established the following:

Finding 1 — Repeated authentication failures

Six failed authentication events were recorded against the same account within approximately 90 seconds.

Assessment: Confirmed.

Finding 2 — Consistent source

All six events originated from 192.168.100.137, with KALI recorded as the workstation.

Assessment: Confirmed.

Finding 3 — Network authentication

All events used Logon Type 3.

Assessment: Confirmed.

Finding 4 — Invalid authentication

The events contained:

Status: 0xC000006D

SubStatus: 0xC000006A

The original Windows event text identified the failure as an unknown username or bad password.

Assessment: Confirmed.

Finding 5 — No matching successful authentication

A targeted Event ID 4624 search for LAB\soclab from 192.168.100.137 returned no matching event during the investigation window.

Assessment: No successful authentication observed.

Finding 6 — No evidence of post-authentication activity

The investigation did not identify any evidence of activity occurring after a successful authentication.

Assessment: No post-authentication compromise was observed within the scope of this case.

10. Severity Assessment

Lab Severity: Medium

The severity classification is based on the combination of repeated authentication failures, the targeted domain account, the short time period and the role of the target Windows system.

Factors increasing concern:

- Six authentication failures occurred in a short period.
- All attempts targeted the same account.

- The attempts originated from the same source.
- The target was a domain-connected Windows Server/domain controller.
- The activity was sufficiently repetitive to trigger the configured detection threshold.

Factors limiting the observed impact:

- No matching successful authentication was identified.
- No evidence of post-authentication activity was observed.
- The account was a standard domain user in the lab.
- The activity was intentionally generated as a controlled simulation.

The severity therefore represents the **risk level assigned to the observed event pattern**, not a claim that a compromise occurred.

11. Response Actions

For a production environment, the following response sequence would be appropriate:

Immediate

1. Validate the source IP and identify the originating endpoint.
2. Confirm whether the targeted account owner initiated the authentication attempts.
3. Check whether the account is privileged, service-related or a normal user account.
4. Search for successful Event ID 4624 activity from the same source.
5. Check for additional authentication failures involving other accounts.
6. Review surrounding security events for signs of follow-on activity.

If malicious activity is confirmed

1. Temporarily contain the source endpoint according to organisational procedures.
2. Consider blocking or restricting the source where appropriate.
3. Protect the targeted account according to incident-response policy.
4. Review authentication activity across other domain resources.
5. Search for additional indicators from the same source.

6. Preserve relevant logs and evidence.
7. Escalate to the appropriate incident-response tier if compromise is suspected.

Important operational consideration

Account lockout should be implemented carefully. Excessively aggressive lockout thresholds can themselves create a denial-of-service condition. MITRE identifies account-use policies and lockout controls as a mitigation for brute-force activity while noting this trade-off.

12. Recommended Detection Improvement

The lab detection rule successfully identified the controlled activity, but a production SOC would benefit from additional correlation.

A stronger analytic could correlate:

Multiple 4625 failures

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Same source IP

↓

Same account

↓

Short time window

↓

4624 success from same source

↓

Post-authentication activity

The highest-priority condition would be a successful authentication following multiple failures.

Additional useful correlations would include:

- Multiple targeted accounts from one source.

- Authentication failures against privileged accounts.
- Authentication attempts from unusual endpoints.
- Repeated failures followed by a successful login.
- Authentication activity outside normal user patterns.
- Correlation with endpoint security alerts.
- Correlation with SMB, RDP or other remote-service activity.

This would reduce false positives while increasing confidence when a failed-authentication sequence is followed by successful access.

13. Analyst Conclusion

The investigation identified a concentrated sequence of six failed network authentication attempts against the LAB\soclab account from the Kali host at 192.168.100.137.

The Windows Security logs consistently recorded Event ID 4625, Logon Type 3 and NTLM authentication. The events occurred within approximately 90 seconds and were sufficient to trigger the configured multiple-failure detection rule.

The observed behaviour is consistent with password-guessing activity and maps to MITRE ATT&CK T1110.001.

However, the investigation did not identify a corresponding successful Event ID 4624 from the same source and target account. Consequently, the evidence collected during this investigation does not demonstrate successful account compromise.

The case is therefore closed as a **controlled password-guessing simulation with no observed successful authentication**.

14. Evidence Preserved

The following files were generated during the investigation:

incident-001-evidence.csv

incident-001-detailed-evidence.csv

The detailed evidence file contains the extracted Windows Security Event fields used to support the investigation.

Recommended evidence directory:

evidence/

├─ incident-001-evidence.csv

└─ incident-001-detailed-evidence.csv

15. Analyst Notes

This case was intentionally created as a controlled SOC laboratory exercise.

The purpose of the exercise was to demonstrate the complete SOC workflow:

Authentication Activity

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Detection

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Alert

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Triage

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Evidence Collection

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Source / Account Correlation

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Timeline Construction

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Successful Login Check

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MITRE ATT&CK Mapping

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Severity Assessment



Response Decision



Incident Closure

No production environment or real user account was involved.